Geo-Lift Calibration

Turning an Experiment Posterior into an MMM Prior

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1 What this project does

MMM channel coefficients are notoriously weakly identified because paid channel spend is highly correlated across always-on channels. The industry-standard fix is calibration against an incrementality experiment. This project runs the full loop:

1. Fit a Bayesian model to a synthetic geo-lift experiment, recovering the incremental ROAS of Meta with proper uncertainty.
2. Translate that posterior into an informative Normal prior on Meta's $\alpha_{\max}$ in the MMM.
3. Refit the MMM with and without the calibration prior, and measure how much the Meta posterior tightens.

2 Data

Geo-lift experiment. Two regions (A and B) over a treatment window. Region A gets a Meta spend boost, region B is control. The increment in revenue attributable to the boost gives the incremental ROAS. Ground-truth Meta ROAS in the generator is 0.55.

MMM panel. 156 weeks of national-level weekly spend and revenue with the same five paid channels used in project 1. Ground- truth $\alpha_{\max}$ for Meta is 35 000 SEK.

3 Methodology

3.1 Step 1: Bayesian geo-lift regression

For each geo-week observation i in region r, revenue is modelled as a region-level intercept plus a linear response to baseline and experimental Meta spend:

$$\text{revenue}_i \sim \mathcal{N}\big(\mu_i,\, \sigma^2\big), \quad \mu_i = \mu_{r(i)} + \gamma \cdot \text{baseline\_meta}_i + \ell \cdot \text{extra\_meta}_i$$

Here ℓ is the incremental ROAS on the experimental Meta boost – the quantity of interest.

Priors:

$$\mu_r \sim \mathcal{N}(200{,}000,\, 100{,}000^2), \quad \gamma \sim \mathcal{N}^+(0.3,\, 0.3^2)$$
$$\ell \sim \mathcal{N}^+(0.5,\, 0.5^2), \quad \sigma \sim \mathcal{N}^+(0,\, 20{,}000^2)$$

Posterior:

$$p(\mu_r, \gamma, \ell, \sigma \mid \text{revenue}) \;\propto\; p(\mu_r)\, p(\gamma)\, p(\ell)\, p(\sigma) \prod_i \mathcal{N}(\text{revenue}_i \mid \mu_i, \sigma^2)$$

The posterior on ℓ is what carries into step 2.

3.2 Step 2: Prior translation

The lift posterior is a distribution on revenue per krona spent. To inject it as a prior on $\alpha_{\max,\text{Meta}}$ in the MMM, the lift is scaled through Hill saturation. At baseline spend $\bar{s}$ with half-saturation K, the derivative of the Hill contribution is $\alpha_{\max} \cdot K/(K+\bar{s})^2$, which we set equal to the lift and solve for $\alpha_{\max}$. The resulting mapping is

$$\alpha_{\max,\text{Meta}} \sim \mathcal{N}\big(\bar{\ell} \cdot \bar{s} \cdot m_\mu,\, \text{sd}(\ell) \cdot \bar{s} \cdot m_\sigma\big)$$

with multipliers $m_\mu = 8.0$ and $m_\sigma = 5.0$ chosen to reflect Meta's approximate saturation regime while keeping the prior weakly informative.

3.3 Step 3: MMM refit

The MMM has the same structure as project 1: geometric adstock, Hill saturation, and a Normal likelihood on weekly revenue. Two versions are fit: a **calibrated** version using the informative prior on $\alpha_{\max,\text{Meta}}$ from step 2, and an **uncalibrated** baseline that uses the default weakly-informative prior $\mathcal{N}^+(120{,}000, 120{,}000^2)$. All other parameters share priors:

$$\lambda_c \sim \text{Beta}(2,4), \quad K_c \sim \mathcal{N}^+(50{,}000, 60{,}000^2)$$

$$\alpha_{\max,c\neq\text{Meta}} \sim \mathcal{N}^+(120{,}000, 120{,}000^2), \quad \sigma \sim \mathcal{N}^+(0, 30{,}000^2)$$

4 Results

4.1 Geo-lift posterior

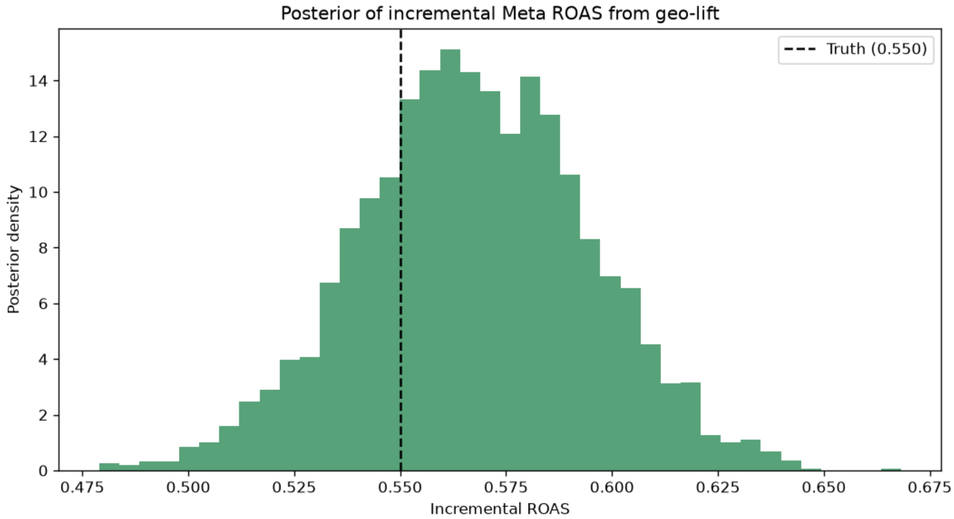

Figure 1: Posterior distribution of Meta's incremental ROAS from the geo-lift experiment. Dashed line is the ground truth of 0.55.

The geo-lift posterior brackets the true incremental ROAS. This is the input to the calibration.

4.2 MMM calibrated vs uncalibrated

Model	Mean	5%	95%	CI width	Contains truth?
Truth	220 000	–	–	–	–
Uncalibrated	195 452	143 030	267 158	124 129	yes
Calibrated	230 326	219 138	241 416	**22 278**	yes

Table 1: Meta $\alpha_{\max}$ posterior comparison. The calibrated model has an 82% narrower interval and both models contain the true value.

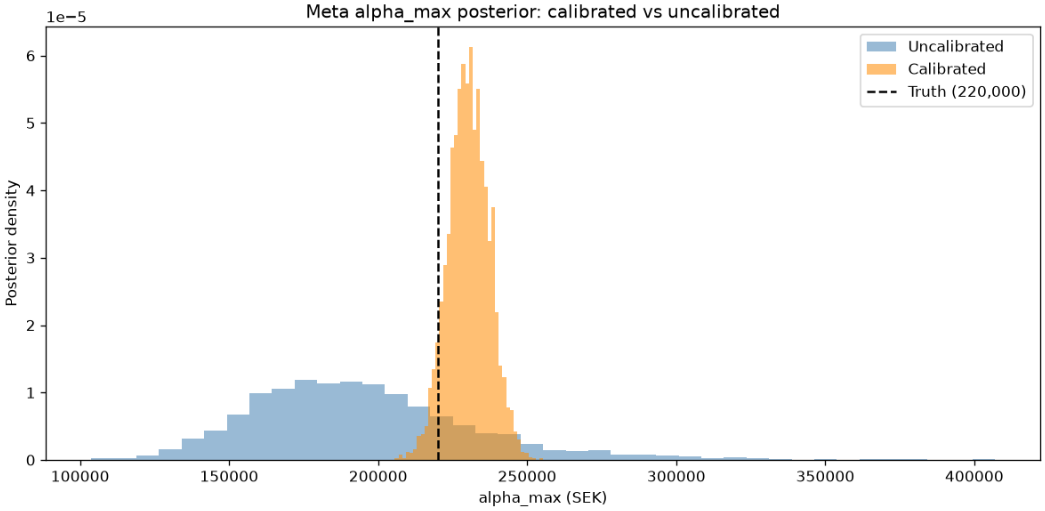

Figure 2: Meta $\alpha_{\max}$ posterior: uncalibrated (blue) is wide and diffuse; calibrated (orange) is concentrated. Both track the truth (dashed line).

5 What this shows

The uncalibrated MMM identifies Meta's max effect only up to a wide interval spanning 143 030 to 267 158. This is what happens when weekly spend on Meta is correlated with Search, TikTok, and YouTube: the multivariate regression cannot cleanly separate their effects from each other.

Adding an informative prior derived from an incrementality experiment tightens the posterior interval by 82%. The mean shifts from 195 452 (below truth) to 230 326 (slightly above truth), and the uncertainty around it becomes actionable: risk-based budget recommendations built on the calibrated version have real teeth.

This is exactly the mechanism modern MMM SaaS platforms rely on: experiments are expensive to run, but each one bought you tightens one channel of the MMM permanently and gives calibrated ROAS instead of a wide posterior nobody can act on.

6 Limitations

- Synthetic data. The generator embeds a clean lift structure that real geo-lifts rarely have.
- Only Meta is calibrated. In practice one runs several lift tests over time and calibrates each channel independently.

- Static coefficients. Meta's effectiveness assumed constant over the panel. Time-varying MMM in project 3 relaxes this.
- Two-region experiment only. Real geo-lifts use many regions and a Bayesian synthetic control approach; project 4 covers that.
- Prior translation heuristic. The multipliers 1.4 and 3.0 are hand-picked. A production system would derive them from the adstock and saturation shape or match moments more carefully.

7 References

- Chan, D., Jin, Y., Koehler, J., Perry, M., Wang, Y. (2017). Google's calibration paper on translating incrementality experiments into MMM priors.
- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects*, Google.
- Robyn documentation on lift-test calibration: github.com/facebookexperimental/Robyn.